

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
B.Tech Degree S6 (S, FE) Examination January 2024 (2019 Scheme)

Course Code: ADT 302
Course Name: CONCEPTS IN BIG DATA ANALYTICS

Max. Marks: 100

Duration: 3 Hours

PART A

Answer all questions, each carries 3 marks.

Marks

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| 1 | List the different characteristics of Big data. | (3) |
| 2 | What is Big data Platform? What are its characteristics? | (3) |
| 3 | What are the benefits of stream processing? | (3) |
| 4 | Illustrate the stream data architecture. | (3) |
| 5 | What are the functions of datanode and namenode? | (3) |
| 6 | List the different Hadoop commands for basic file system operations. | (3) |
| 7 | What are the various Execution Modes of Pig? | (3) |
| 8 | List the various DML commands supported by HBase. | (3) |
| 9 | Discuss the general list operations in R with example. | (3) |
| 10 | Discuss the features of R Language. | (3) |

PART B

Answer one question from each module, each carries 14 marks.

Module I

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| 11 | a) Explain in detail about Nature of Data and its types. | (7) |
| | b) Illustrate Big Data Architecture. | (7) |

OR

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| 12 | a) Explain in detail the benefits of Intelligent Data Analysis. | (7) |
| | b) Discuss the importance of Big Data. | (7) |

Module II

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| 13 | a) Explain Sampling techniques for efficient stream processing. | (7) |
| | b) Explain DGIM algorithm with an example. | (7) |

OR

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| 14 | a) Explain Bloom filter with an example. | (7) |
| | b) How to use Flajolet- Martin algorithm to count the distinct elements in a stream? | (7) |

Module III

- 15 a) Write about the anatomy of file write operation in HDFS. (7)
b) Describe Hadoop architecture with neat sketch. (7)

OR

- 16 a) Explain the Map reduce execution pipeline. (7)
b) Explain in detail Hadoop Ecosystem and its Core Components. (7)

Module IV

- 17 a) Explain the Architecture of Hive. (7)
b) Compare HBase with RDBMS. (7)

OR

- 18 a) What is HBase and what are its features? Explain HBase Data Model. (7)
b) List out the different data processing operations in Pig. (7)

Module V

- 19 a) Construct a data frame using R and perform different operations on it. (7)
b) Illustrate the operations performed in R vectors. (7)

OR

- 20 a) List the applications of R programming. (7)
b) Write an R program to find the factorial of a number by reading the input from a file and check whether it is palindrome. (7)
